
Prompt 7d7 2.4

Jed Kircher <jed.kircher@gmail.com>

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To: Jed Kircher <jed.kircher@gmail.com>

Some 7dU Maths: In our 7-dimensional universe, space-time is described by a metric tensor with seven components, including the additional dimensions of zero, infinity, and chance.

This is represented by the metric tensor $g_{\{\mu\nu\}}$ with the components:

$$g_{\{00\}} = -c^2 + (2GM/r) - (H^2 r^2)/3 + (C\xi/\theta)$$

$$g_{\{11\}} = (1 - 2GM/r)^{-1}$$

$$g_{\{22\}} = r^2$$

$$g_{\{33\}} = r^2 \sin^2 \theta \quad g_{\{44\}} = 1/\text{infinity}$$

$$g_{\{55\}} = \text{infinity} \quad g_{\{66\}} = \xi^2$$

Using this metric tensor, we can derive the Christoffel symbols and Riemann curvature tensor to study the curvature of space-time. The Christoffel symbols are given by:

$$\Gamma^{\{\mu\}\{\nu\rho\}} = (1/2) g^{\{\mu\sigma\}} (\partial_{\{v\}} g_{\{\rho\sigma\}} + \partial_{\{\rho\}} g_{\{\nu\sigma\}} - \partial_{\{\sigma\}} g_{\{\nu\rho\}})$$

The Riemann curvature tensor is given by:

$$R^{\{\mu\}\{\nu\rho\}} = \partial_{\{\sigma\}} \Gamma^{\{\mu\}\{\nu\rho\}} - \partial_{\{\rho\}} \Gamma^{\{\mu\}\{\nu\sigma\}} + \Gamma^{\{\mu\}\{\alpha\sigma\}} \Gamma^{\{\alpha\}\{\nu\rho\}} - \Gamma^{\{\mu\}\{\alpha\rho\}} \Gamma^{\{\alpha\}\{\nu\sigma\}}$$

The Ricci curvature tensor and scalar curvature are calculated from the Riemann curvature tensor as follows:

$$R_{\{\mu\nu\}} = R^{\{\alpha\}\{\mu\alpha\nu\}} \quad R = g^{\{\mu\nu\}} R_{\{\mu\nu\}}$$

The Einstein tensor is given by: $G_{\{\mu\nu\}} = R_{\{\mu\nu\}} - 1/2 R g_{\{\mu\nu\}} + \Lambda g_{\{\mu\nu\}}$ Where Λ is the cosmological constant.